

AKSHAT MEHTA

Generative AI Engineer – LLMs, RAG & Machine Learning
Open to Relocate

Rochester, NY | Portfolio | akshat2404@gmail.com | LinkedIn/akshatmehta24 | Github/akshat240401

EDUCATION

Rochester Institute of Technology

Master of Science in Computer Science, Adv. Cert. Artificial Intelligence

Aug 2023 - May 2026

Rochester, NY

University of Mumbai

Bachelor of Engineering in Computer Engineering

Jul 2019 - May 2023

Mumbai, India

SKILLS

Programming: Python, SQL

Generative AI: LLMs, RAG, Prompt Engineering, OpenAI APIs, Azure OpenAI, Function Calling, Structured Outputs

AI/ML: Machine Learning, Deep Learning, PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn, LLM Fine-Tuning

Retrieval: Vector Embeddings, Vector Databases, Semantic Search

Data: PostgreSQL, PySpark, pandas, NumPy, Apache Airflow, Data Pipelines, Data Ingestion, ETL Pipelines, Data Validation

Cloud & MLOps: AWS (SageMaker, EC2, S3, Lambda, Batch), Azure, Docker, MLflow, W&B

ML Systems: Model Development, Model Architectures, Training Pipelines, Inference Pipelines, Model Monitoring

Evaluation: Model Evaluation, Model Validation, Benchmarking, Robustness Testing, Error Analysis, Confidence Calibration

PROJECTS

Live Race Engineer – AI Decision Platform | RAG, Vector Embeddings, Function Calling

May 2026 – Present

- Engineered a **Python data pipeline** to ingest, validate, and process real-time telemetry at **30 Hz**.
- Built a **function-calling LLM workflow** with RAG, vector embeddings, and prompt engineering for grounded decisions.
- Used **structured outputs and retrieval logic** to generate contextual recommendations from telemetry and incidents.
- Added **monitoring, failure simulation, and state recovery** to evaluate reliability across real-time workflows.

Image-Caption Alignment & Refinement | PyTorch, Model Evaluation, Calibration

Nov 2025 – Apr 2026

- Built a **PyTorch inference pipeline** with CLIP embedding caching, batch processing, and automated model evaluation.
- Trained and benchmarked a **CLIP + MLP deep learning model**, achieving **0.714 Macro-F1** and a **9.5% gain**.
- Reduced **Expected Calibration Error by 55%** through robustness testing, error analysis, and confidence calibration.
- Integrated **BLIP, T5, and OWL-ViT** with output validation and fallback routing for reliable model refinement.

Multimodal Note-Taking Assistant | Fine-Tuning, PyTorch, Data Pipelines

Jan 2025 – May 2025

- Built a multimodal **data pipeline** across **15,599 frames and 78 lectures** using vision, speech, and text processing.
- Automated **data ingestion, preprocessing, and validation** for noisy image, audio, and text inputs.
- Fine-tuned and evaluated **ResNet-34 with PyTorch**, achieving **91% accuracy** through cross-validation.
- Fine-tuned **BART-large on 280K+ examples**, achieving **41.1 ROUGE-1** for transformer-based summarization.

AI News Monitor & Weekly Digest | LLMs, Data Ingestion, Structured Outputs

Jul 2026

- Built an automated **data ingestion pipeline** to collect and process emerging AI technology news from RSS feeds.
- Transformed **RSS and XML data into structured JSON** with filtering, deduplication, and date validation.
- Ranked and selected up to **20 relevant articles per run** to prioritize high-value technology updates.
- Integrated an **LLM with structured outputs** to generate grounded summaries and formatted weekly HTML reports.

WORK EXPERIENCE

Full Stack Engineer | Python, Data Pipelines, AWS

University of Mumbai (Shah & Anchor Kutchhi Engineering College)

Sep 2022 – Jul 2023

Mumbai, India

- Built **Python and AWS data pipelines** for candidate-drug research across large molecular datasets.
- Collaborated with researchers to translate analytical requirements into reproducible **research workflows**.
- Reduced workflow runtime by **95.2%** and documented methods and results in a **Research Square preprint**.

Full Stack Developer Intern | Data Validation, API Integration, Testing

Corizo

Jul 2022 – Sep 2022

Noida, India

- Built **REST API integrations** with JSON processing and data validation, reducing response time by **50%**.
- Collaborated with **product and engineering teams** on requirements, code reviews, testing, and iterative releases.
- Applied **automated testing and error handling** to validate integrations and improve release reliability.